

Realization of Quantum Antenna Architectures for Radar Systems Using Entangled Electromagnetic Fields

Kumar Gautam^{1*}, Harish Parthasarathy², Namisha Gupta³ and Divya Punia⁵

Abstract—This work describes a quantum antenna design and approach for radar applications that use quantum electromagnetic fields and superconducting fermionic systems. The system generates high-frequency Josephson currents using Cooper pair interactions within a superconducting cavity, which serves as a quantum radiation source. Photon generation and annihilation operators are used to represent the transmitted electromagnetic field, which interacts with targets to produce reflected quantum signals with Doppler information and range. Quantum cross-correlation techniques are used to estimate the values of the objective parameters. Quantum teleportation and other entanglement-based methods improve detection sensitivity and resistance to noise. The research provides a scalable architecture for next-generation radar systems that can identify many targets simultaneously. This technology has significant advantages over classical radar, including increased sensitivity, interference resistance, and the capacity to detect low-observable targets. To improve resistance to interference, noise, and signal loss, the technique employs entanglement-assisted protocols based on quantum teleportation and channel analysis. This design is scalable and adaptable to real-world deployment, allowing several targets to be monitored simultaneously. The proposed method improves sensitivity, resists jamming, and can identify low-observable items. It eliminates the need for cryogenic cooling at the receiver while leveraging strong light-matter interactions. In this paper, we present a design for compact, high-performance quantum radar systems with potential applications in aerospace, defense, and remote sensing. This will put quantum antennas in line with their classical counterparts. We also describe how to implement quantum antennas in various topologies using materials such as plasmonic metals, carbon nanotubes, and quantum dots.

Index Terms—Entangled electromagnetic fields, microwave Rydberg receiver, Josephson current, Quantum cross-correlation, quantum teleportation protocol and microwave cQED.

I. INTRODUCTION

RADAR technology has been a cornerstone of remote sensing, surveillance, and defense systems for nearly a century. Classical radar systems rely on the transmission of electromagnetic waves and the detection of their echoes to estimate target range, velocity, and bearing. However, conventional radars face fundamental limitations in challenging environments, including strong thermal noise, deliberate jamming, clutter, and low-observable (stealth) targets designed to minimize radar cross-section. These constraints motivate

the exploration of quantum-enhanced radar architectures that exploit non-classical resources such as entanglement and quantum correlations to achieve superior performance. Quantum illumination, originally proposed in the optical domain and later extended to microwaves, offers a promising pathway by using entangled probe states. In the canonical QI protocol, one mode of an entangled pair illuminates the target while the other (idler) is retained locally. Joint quantum measurements on the returned signal and idler can outperform optimal classical strategies, particularly in the low-signal, high-noise regime, even when entanglement is broken by the lossy channel. This paper presents a novel realization of quantum antenna architectures for radar systems based on entangled electromagnetic fields generated via superconducting fermionic systems. We propose a unified framework that combines high-frequency Josephson currents in superconducting cavities as a quantum radiation source with entanglement-based protocols, including elements inspired by quantum teleportation, for enhanced parameter estimation. The approach generalizes classical range-Doppler cross-correlation techniques to quantum operator correlations while introducing a superconductivity-driven antenna cavity for practical generation of the required quantum fields.

Unlike standard quantum illumination protocols that primarily rely on two-mode squeezed vacuum states generated by parametric amplifiers, our framework introduces a superconducting fermionic antenna cavity where Cooper pair interactions directly drive the quantum radiation via high-frequency Josephson currents. This provides a more integrated “quantum antenna” source. We further generalize classical radar cross-correlation to full quantum operator correlations and propose an entanglement-based protocol that interprets target reflection as a channel-dependent distortion in a teleportation-like bidirectional process. This allows simultaneous extraction of range, velocity, and bearing information from the distortion of teleported states, offering a new perspective on quantum channel estimation for radar. The architecture is designed to be scalable and leverages the natural high-frequency quantum effects ($\propto 1/\hbar$) inherent to superconducting systems.

The major contributions of this work include:

- 1) A microscopic derivation of the radiating Josephson current from a fermionic action incorporating Cooper pair fields and controlled electromagnetic interactions.
- 2) Generalization of classical radar signal processing to quantum cross-correlations of photon creation/annihilation operators for multi-target range and velocity estimation.

^{1*,2,3}Quantum Material and Device Vertical, Egreen Quanta, New Delhi, India.

³Department of Electronics and Communications, Netaji Subhas University of Technology, Delhi, India.

- 3) An entanglement-assisted radar protocol that leverages shared entangled states and bidirectional teleportation-like processes to encode and extract target parameters (range, bearing, and radial velocity) with improved sensitivity and noise robustness.
- 4) A scalable superconducting cavity architecture suitable for next-generation quantum radar systems.

The physical realization of this paper relies on superconducting circuits operating at millikelvin temperatures, where Cooper pair tunneling and Josephson parametric processes naturally generate entangled microwave fields. The antenna cavity integrates fermionic modes with a quantum electromagnetic source, enabling high-frequency radiation driven by quantum mechanical effects proportional to ($\propto 1/\hbar$). This setup is compatible with existing circuit quantum electrodynamics (cQED) platforms, Josephson parametric amplifiers (JPAs), and Josephson traveling-wave parametric amplifiers (JTW-PAs), which have already demonstrated microwave entanglement generation in laboratory prototypes. The military (stealth target detection, jamming-resistant surveillance), the aerospace industry, autonomous vehicles, and possibly even short-range biomedical imaging could all benefit from this technology. The main benefits compared to traditional radar include the following: increased sensitivity for detection in highly reflective thermal environments, resistance to interference and jamming, improved performance against targets with poor observability, and the possibility of achieving multi-target resolution while reducing transmitted power. Although propagation losses make long-range operation difficult, the suggested design paves the way for a quantum-enhanced radar that can be used in real-world noisy settings with reduced error exponents. In a superconducting cavity, regulated electromagnetic fields and fermionic Cooper pair dynamics are used directly to generate radiation. Thoroughly expanding range-Doppler estimate with quantum cross-correlations $\text{Tr}(\rho G(t + \tau)F(t))$. Implementing robust multi-parameter sensing through the utilization of shared entanglement and reverse-teleportation techniques. Support for multi-target scenarios, reduced detectable transmit power, enhanced sensitivity in noisy conditions, and resistance to jamming are all desirable qualities. This strategy fills in the gaps in practical antenna integration and full radar capabilities by utilizing established superconducting cQED technology. This strategy fills in the gaps in practical antenna integration and full radar capabilities by utilizing established superconducting cQED technology, providing a comprehensive solution for advanced radar systems that require high performance in challenging environments. Additionally, the use of superconducting cQED technology allows for improved efficiency and reliability in radar operations, making it an ideal choice for next-generation radar applications.

The remainder of the paper is organized as follows: Section II reviews background and prior art. Section III presents the mathematical formulation. Subsequent sections detail the superconducting quantum antenna and the entanglement-based radar protocol.

II. BACKGROUND AND PRIOR ART

The concept of exploiting quantum resources for target detection traces back to the early 2000s. Quantum illumination (QI) was formalized by Lloyd (2008) and Tan et al. (2008) in the optical regime using entangled photon pairs, demonstrating a potential quadratic improvement in error probability for detecting a weakly reflecting target in thermal noise. The protocol was later adapted to microwaves by Guha and Erkmen (2015) and others, recognizing that the microwave regime with its naturally stronger thermal noise offers a more favorable setting for observing quantum advantage. Experimental progress accelerated with superconducting circuit platforms. Key demonstrations include the 2019–2020 microwave QI experiments by Barzanjeh et al., which used a Josephson parametric converter (JPC) to generate entangled signal-idler modes at millikelvin temperatures. The signal illuminated a room-temperature target at 1 meter distance, while the idler was retained for joint (often digital heterodyne) measurement, showing improved signal-to-noise performance compared to certain classical benchmarks. Subsequent works explored Josephson traveling-wave parametric amplifiers (JTW-PAs) for broader bandwidth entangled sources and cryogenic end-to-end setups achieving modest error-exponent advantages (0.8 dB) over theoretical classical limits. Other contributions include quantum two-mode squeezed radar prototypes and studies on receiver designs (e.g., phase-conjugate or optical parametric amplifier receivers). Despite these advances, most experimental demonstrations remain limited to short ranges (1 m in free space, up to tens of meters in controlled delay-line setups) due to severe propagation losses, the need for cryogenics, and the difficulty of preserving useful quantum correlations over long distances. True long-range quantum radar with clear advantage over optimally engineered classical systems in realistic outdoor conditions has not yet been realized. Moreover, many implementations focus on detection (hypothesis testing: target present/absent) rather than full parameter estimation (range, Doppler, bearing) for multiple dynamic targets.

Prerequisites: Successful implementation requires millikelvin cryogenic environments for entanglement generation and preservation, high-efficiency coupling between the superconducting cavity and free-space propagating modes, low-loss quantum-limited amplification, and advanced quantum receivers capable of joint measurements. Progress in circuit QED, impedance matching for quantum antennas, and hybrid quantum-classical signal processing will be essential. While current technology supports proof-of-principle short-range systems, scaling to practical radar ranges demands breakthroughs in mitigating propagation losses and thermal noise.

III. MATHEMATICAL FORMULATION:

At the initial time $t = 0$, the quantum electromagnetic field or equivalently, the photon state at the ground station acting as the transmitter is described by the density matrix $\rho_T(0)$. In a similar manner, the quantum state associated with the receiver, which corresponds to the target, is represented by $\rho_R(0)$. These states may be realized as coherent photon states,

photon number (Fock) states, or arbitrary linear superpositions of such states. A key assumption at $t = 0$ is that the photon states of the transmitter and the receiver are not entangled. Consequently, the joint quantum state of the transmitter (Tx) and receiver (Rx) can be expressed as the tensor product

$$\rho_T(0) \otimes \rho_R(0)$$

which explicitly reflects their statistical independence at the initial moment. Let X_T denote the electromagnetic field operator associated with the transmitter and X_R the corresponding field operator of the receiver. These fields act on distinct tensor factors of the composite Hilbert (or Fock) space formed by the tensor product of the transmitter and receiver subsystems, ensuring that their quantum degrees of freedom are initially separable.

In conventional classical radar systems, consider a target receiver Rx located at a distance d from the transmitter Tx. If $x(t)$ denotes the electromagnetic (photon) signal transmitted by Tx, then the signal received back at Tx after reflection from Rx can be modeled as

$$y(t) = x(t - 2d/c) + n(t)$$

where c is the speed of light and $n(t)$ represents additive channel noise. The range information is extracted by computing the cross correlation between the transmitted and received signals at the transmitter, defined as

$$R_{yx}(\tau) = \langle y(t + \tau) \cdot x(t) \rangle$$

Substituting for $y(t)$ yields

$$R_{yx}(\tau) = \langle y(t + \tau) \cdot x(t) \rangle = \langle x(t + \tau - 2d/c) \cdot x(t) \rangle = R_{xx}(\tau - 2d/c)$$

where $R_{xx}()$ is the autocorrelation function of the transmitted signal. The cross correlation attains its maximum magnitude at $\tau = 2d/c$, thus enabling direct estimation of the target range d . This principle forms the foundation of range estimation in classical radar systems. To additionally estimate the velocity of a moving target Rx, Doppler frequency shifts induced by the target's motion must be incorporated into the signal model and subsequent correlation analysis.

Specifically, suppose v denotes the component of the target velocity along the inward radial direction from the receiver Rx toward the transmitter Tx. If the electromagnetic signal transmitted from Tx is denoted by $x(t)$, it may be represented in the frequency domain via the Fourier decomposition

$$x(t) = \int X(\omega) \cdot \exp(j\omega t) d\omega / 2\pi$$

After reflection from the moving target Rx, the received signal at the transmitter incorporates both the propagation delay associated with the target range and the Doppler shift induced by the target's motion. The received signal may therefore be written as

$$y(t) = \int X(\omega) \cdot \exp(j\omega(1 + v/c)(t - d/c)) d\omega + n(t) = x((1 + v/c)(t - 2d/c)) + n(t)$$

where $n(t)$ denotes additive channel noise. The cross correlation between the transmitted and received signals is then given by

$$\begin{aligned} R_{yx}(t, \tau) &= \langle y(t + \tau) \cdot x(t) \rangle = \langle x((1 + v/c)(t + \tau - 2d/c)) \cdot x(t) \rangle \\ &= R_{xx}((1 + v/c)(t + \tau - 2d/c) - t) = R_{xx}(vt/c + (1 + v/c)(\tau - 2d/c)) \end{aligned}$$

where $R_{xx}()$ denotes the autocorrelation function of the transmitted signal. By analyzing this cross correlation as a joint function of the target range t and τ , the target range d and radial velocity v can be estimated. In particular, for a fixed observation time t , plotting $R_{yx}(t, \tau)$ as a function of τ reveals both the time delay $2d/c$ and the Doppler-induced time scaling factor $1 + v/c$, thereby enabling simultaneous estimation of the target's range and velocity.

The function gets compressed by a factor of $1 + v/c$ and delayed by $2d/c$. More generally, consider several targets in the sky located at positions $R_1, \dots, R_p$ and moving with velocity vectors $V_1, \dots, V_p$ respectively. Let $x(t)$ be the signal sent out by Tx in the form of a spherical wave. Assume that Tx is located at r_0 . Then, the received signal after getting reflected by these targets is given by the linear superposition:

$$y(t) = \sum_{k=1}^p a(k) x((1 + V_k/c)(t - 2|R_k - r_0|/c)) + n(t)$$

where $a(k)$ is the attenuation of the signal along the straight line path of length $2|R_k - r_0|$ joining Tx and Rx back and forth. If we take into account the fact that spherical electromagnetic waves travel as $x(t - r/c)/r$ where r is the radial distance, then this formula must be modified to

$$y(t|r_0) = \sum_{k=1}^p a(k) x((1 + V_k \cdot (R_k - r_0)/c |R_k - r_0|)(t - 2|R_k - r_0|/c)) / |R_k - r_0| + n(t)$$

From the received process $y(t)$, $t \geq 0$, we can apply the maximum likelihood or least square method to estimate $\{R_k, V_k\}_{k=1}^p$ from measurements of the signal field $y(t|r_0)$, $t \geq 0$, $r_0 \in D$, where D is a set of spatial positions where we can place our Tx antenna.

The correlation function is therefore compressed by a factor of $1 + v/c$ and delayed by $2d/c$. More generally, consider p targets located at positions $R_1, R_2, \dots, R_p$ and moving with velocity vectors $V_1, V_2, \dots, V_p$, respectively. Let $x(t)$ denote the signal transmitted by Tx, radiated as a spherical electromagnetic wave, and assume that the transmitter is located at position r_0 . The received signal at the transmitter after reflection from all targets can then be modeled as a linear superposition:

$$y(t) = \sum_{k=1}^p a(k) x((1 + V_k/c)(t - 2|R_k - r_0|/c)) + n(t)$$

where $a(k)$ represents the attenuation along the round-trip propagation path of length $2|R_k - r_0|$, and $n(t)$ denotes additive channel noise. Taking into account the fact that a spherical electromagnetic wave propagates as $x(t - r/c)/r$,

where r is the radial distance from the source, this expression must be refined to

$$y(t|r_0) = \sum_{k=1}^p a(k)x((1 + V_k \cdot (R_k - r_0)/c)|R_k - r_0|) \\ (t - 2|R_k - r_0|/c))/|R_k - r_0| + n(t)$$

Here, the Doppler factor depends on the radial component of the target velocity,

$$V_k^{(r)} = \frac{V_k \cdot (R_k - r_0)}{\|R_k - r_0\|}$$

which determines the frequency shift experienced by the reflected signal. From observations of the received process $y(t | r_0)$, $t \geq 0$, one may apply standard maximum likelihood or least-squares estimation techniques to infer the target parameters $\{R_k, V_k\}_{k=1}^p$ using measurements collected at multiple transmitter positions $r_0 \in D$, where D denotes a set of spatial locations over which the transmitting antenna can be deployed. This spatial diversity enables simultaneous estimation of target range, bearing, and velocity for multiple targets.

Physically, the classical received signal may be viewed as a delayed and time-scaled image of the transmitted waveform, where the delay encodes the target range and the compression (or dilation) factor encodes the Doppler shift due to target motion. In deterministic radar, this information is extracted by correlating the reflected signal with a locally stored replica of the transmitted waveform. The environment acts as a classical channel whose parameters—propagation delay, attenuation, and Doppler scaling—are inferred solely from distortions imposed on the signal. This perspective naturally generalizes to the quantum domain. In entanglement-based sensing, the transmitted field is no longer treated as an isolated classical waveform but as one subsystem of a larger quantum state that is entangled with a reference (idler) field retained at the transmitter. The target and propagation medium then act as a quantum channel acting on the signal subsystem. Range, velocity, and angular information are encoded not only in temporal delays and frequency shifts but more fundamentally in the way the channel modifies quantum correlations between the signal and idler. Even when environmental noise destroys entanglement during propagation, residual quantum correlations can survive and provide enhanced parameter sensitivity when joint measurements are performed. From this viewpoint, classical cross-correlation is replaced by quantum operator correlations, and delay and Doppler estimation becomes a problem of quantum channel estimation. The reflected field carries channel-dependent distortions of an initially correlated quantum state, and extracting target information corresponds to optimally decoding these distortions. This conceptual shift—from waveform matching to correlation-preserving channel probing—forms the physical foundation of quantum radar and motivates the use of entanglement, joint measurements, and teleportation-inspired protocols discussed in the sections that follow.

Operationally, classical radar estimation is implemented by measuring the reflected electromagnetic field with time-resolved receivers (e.g., coherent mixers and matched filters) and forming cross-correlation statistics between the detected return signal and a locally stored reference copy of the transmitted waveform. Time of arrival is inferred from the delay that maximizes the correlation output, while target velocity is extracted from the Doppler-induced time scaling or frequency shift measured at the detector. The receiver performs these measurements independently on the returned field, and all inference is based on classical signal statistics derived from the detected intensity and phase. In entanglement-based sensing, the measurement paradigm is fundamentally altered. The transmitted field (signal mode) is generated jointly with a reference field (idler mode) in an entangled quantum state. The signal mode propagates toward the target and is detected after reflection, while the idler mode is retained locally and measured in a synchronized manner. Detectors no longer act independently on the received field alone; instead, joint measurements are performed on the returned signal and the stored idler, using quantum receivers such as joint homodyne/heterodyne detectors, phase-sensitive amplifiers, or correlation-based photon number measurements. From an operational standpoint, the propagation and reflection from the target implement a quantum channel acting on the signal mode. This channel introduces delay, attenuation, Doppler-induced mode transformation, and noise, which collectively modify the joint quantum state shared between the signal and idler. Target parameters are then inferred by measuring suitable observables such as field quadratures, photon number correlations, or higher-order operator moments that are sensitive to these channel-induced distortions. Importantly, the estimation process exploits correlations between the detected signal and idler outcomes, rather than relying solely on properties of the received field. Even in regimes where environmental noise destroys entanglement prior to detection, residual quantum correlations captured through appropriately designed joint measurements can enhance sensitivity compared to purely classical detection strategies. In this sense, range and velocity estimation become problems of quantum channel estimation: the detector architecture and measurement operators are chosen to optimally extract channel parameters by comparing the statistics of joint signal–idler measurement outcomes against the known prepared quantum state. This operational shift from single-mode detection to correlation-based joint measurement provides the practical foundation for quantum radar architectures. It links classical matched filter processing to quantum cross-correlation of operators and motivates the use of superconducting parametric sources, quantum-limited amplifiers, and joint receivers in entanglement-assisted radar systems.

In the entanglement-assisted radar architecture considered here, the electromagnetic field illuminating the target is generated and radiated by a superconducting Josephson-based quantum antenna, rather than by a classical microwave source. The antenna consists of a superconducting cavity incorporating fermionic degrees of freedom, in which Cooper pair tunneling produces high-frequency Josephson currents that act as a

fundamentally quantum radiation source. These currents arise from the coherent dynamics of paired electrons and are intrinsically quantum mechanical, with characteristic frequencies proportional to $1/\hbar$, enabling generation of nonclassical microwave fields at the carrier level. Operationally, the Josephson antenna serves a dual role: it functions simultaneously as a quantum field generator and as a radiating element that couples the internal superconducting modes to free space or guided electromagnetic modes. By embedding the Josephson junction or superconducting cavity within an appropriately matched antenna structure, the internally generated quantum electromagnetic fields described in terms of photon creation and annihilation operators are converted into propagating signal modes that illuminate the target. Unlike classical antennas driven by external oscillators, the radiation process here is directly governed by the microscopic fermionic dynamics encoded in the superconducting action. The quantum antenna is engineered to generate entangled signal-idler field pairs using Josephson parametric processes naturally available in circuit quantum electrodynamics (cQED) platforms. One mode (the signal) is radiated toward the target, while its entangled partner (the idler) is retained locally in a low loss superconducting environment. This joint state preparation is fully compatible with existing Josephson parametric amplifiers and traveling wave parametric devices, which provide broadband, quantum limited generation of entangled microwave fields. From a measurement perspective, the reflected signal field captured by the receiving antenna is not processed in isolation. Instead, it is routed to a joint quantum receiver, where it is detected together with the stored idler field. Superconducting detection chains—comprising phase sensitive amplifiers, quantum limited amplifiers, and homodyne or heterodyne detection stages—enable measurement of joint observables such as field quadrature correlations or photon number correlations. Target induced delay, Doppler shift, and attenuation manifest as distortions of the joint signal-idler statistics rather than merely as changes in received power. In this operational framework, the propagation to and reflection from the target implement a quantum channel acting on the signal mode, while the Josephson quantum antenna provides a stable, phase coherent, and entanglement enabled reference against which these channel induced distortions are measured. Range, velocity, and bearing estimation thus reduce to a problem of quantum channel estimation performed via joint measurements, generalizing classical matched filter radar processing to quantum cross correlation of field operators. The superconducting Josephson antenna therefore forms the physical and operational bridge between microscopic quantum dynamics and macroscopic radar scale sensing. We now wish to generalize these ideas to the case of quantum antennas for estimating the range and bearing of multiple targets. So suppose that to start with, there is only one target located at the position R and moving with an radially inward velocity v . The radially propagating quantum electromagnetic field generated at Tx located at the origin has the form

$$F(t - r/c)/r = \sum_k c(k)e_k(t - r/c)/r + c(k)^*\bar{e}_k(t - r/c)/r$$

where $c(k), c(k)^*, k = 1, 2, \dots$ are single photon annihilation and creation operators satisfying the CCR

$$[c(k), c(m)] = [c(k)^*, c(m)^*] = 0, [c(k), c(m)^*] = \delta(k, m)$$

Here $F(t - r/c)/r$ stands for $[E(t, r), B(t, r)]$, the six component electromagnetic field vector. The received em signal after hitting the target and getting reflected by it, taking into account Doppler shift is given by

$$G(t) = 2d.F((1 + v/c)(t - 2d/c)) = \sum_k [c(k)e_k((1 + v/c)(t - 2d/c)) + c(k)^*\bar{e}_k((1 + v/c)(t - 2d/c))]$$

Now assume that we have prepared our initial em field at Tx to be in the state ρ . Then, the cross correlation between the transmitted em signal and the received em signal in this state would be given by

$$\begin{aligned} R_{GF}(t, \tau) &= \text{Tr}(\rho.(G(t + \tau).F(t))) \\ &= \sum_{k,m} [\text{Tr}(\rho.c(k)c(m))(e_k((1 + v/c)(t + \tau - 2d/c)), e_m(t)) \\ &\quad + \text{Tr}(\rho.c(k).c(m)^*)(e_k((1 + v/c)(t + \tau - 2d/c)), \bar{e}_m(t)) \\ &\quad + \text{Tr}(\rho.c(k)^*c(m)).(\bar{e}_k((1 + v/c)(t + \tau - 2d/c)), e_m(t)) + \\ &\quad \text{Tr}(\rho.c(k)^*c(m)^*)(\bar{e}_k((1 + v/c)(t + \tau - 2d/c)), \bar{e}_m(t))] \end{aligned}$$

and in a similar way, by plotting this cross-correlation function as a function of t, τ , we can estimate d, v .

A. Single-Target Quantum Antenna Model

Consider a single target located at position $\mathbf{R}$ moving with radial velocity v relative to a transmitter located at the origin.

1) Target Range:

$$d = \|\mathbf{R}\|$$

This defines the instantaneous range d as the Euclidean distance from the transmitter to the target. It determines the round-trip propagation delay $2d/c$.

The diagram shows the basic geometry of a single-target radar detection scenario, where a transmitter located at the origin detects a moving target in space. The black point marked Tx at coordinates $((0, 0))$ represents the transmitter or radar antenna. This is the point from which electromagnetic or quantum radar signals are emitted. The horizontal line labeled (x) represents the reference axis of the coordinate system. The red point labeled Target at (R) represents the object being tracked, such as an aircraft, ship, drone, or vehicle. Its location is given by the position vector ($\mathbf{R}$), measured from the transmitter. The dashed line connecting the transmitter to the target indicates the direct line-of-sight path between them. The distance from the transmitter to the target is denoted by:

$$[d = |\mathbf{R}|]$$

where $(|\mathbf{R}|)$ is the magnitude of the position vector. This quantity is the range of the target and determines the signal travel time. The blue arrow near the target indicates the radial

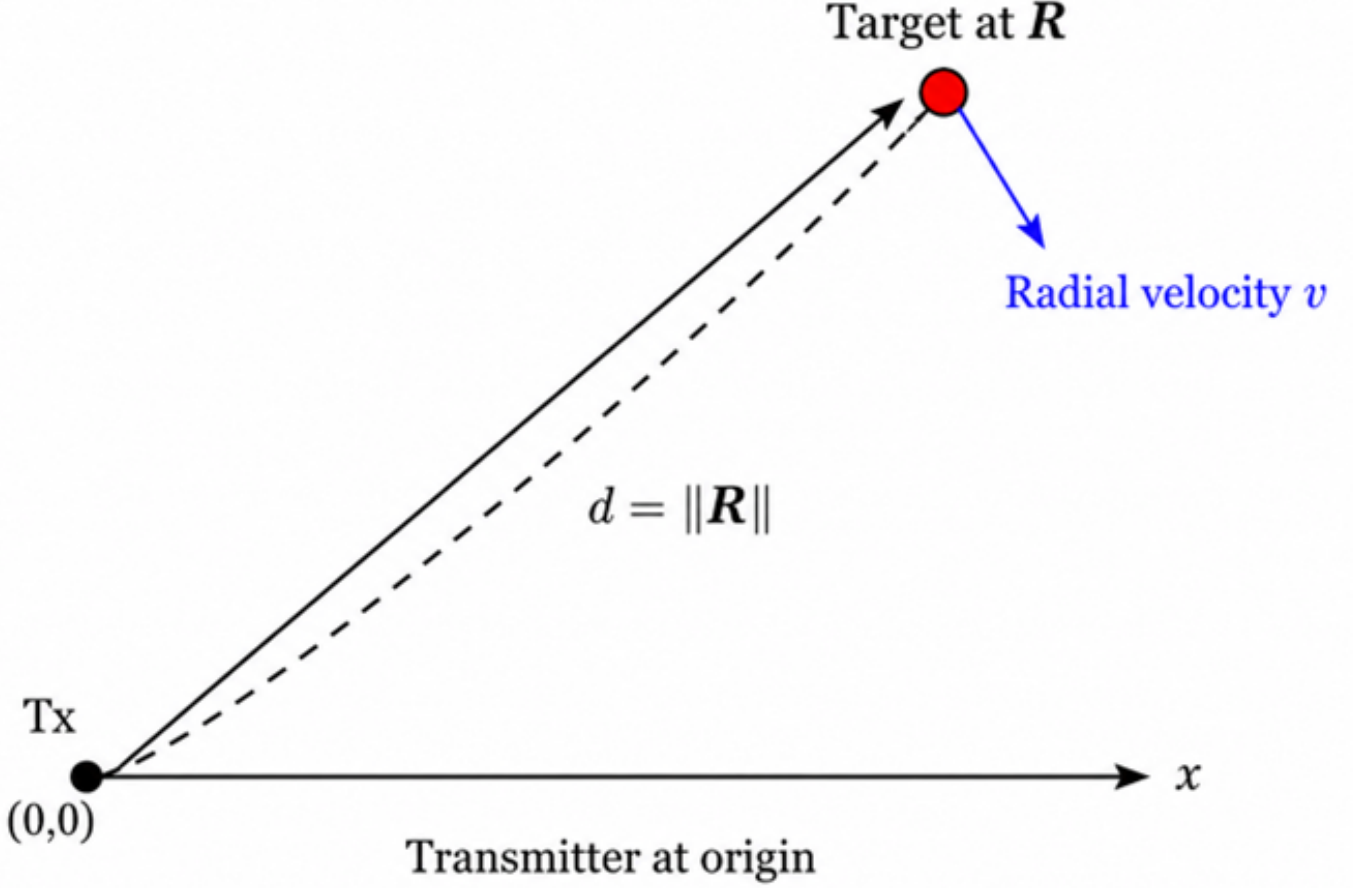


Fig. 1. Schematic of a single-target radar detection scenario

velocity (v), which is the component of the target's motion along the line joining the target and transmitter. Radial velocity is important because it causes a Doppler frequency shift in the returned signal. If the target moves toward the transmitter, the frequency increases; if it moves away, the frequency decreases. Thus, the diagram captures the two key radar observables: range (d) and radial velocity (v), essential for tracking and target estimation.

2) *Outgoing Quantum Electromagnetic Field*: The transmitted quantum field at a distance r from the transmitter is given by the spherical wave:

$$\frac{F(t - r/c)}{r} = \sum_k \left[c_k \frac{e_k(t - r/c)}{r} + c_k^\dagger \frac{\bar{e}_k(t - r/c)}{r} \right]$$

where:

- $F(t - r/c)$ is the field operator evaluated at the retarded time,
- $1/r$ accounts for spherical spreading loss,
- $e_k(t)$ are normalized complex mode waveforms,
- $c_k, c_k^\dagger$ are bosonic annihilation and creation operators.

The mode operators satisfy:

$$[c_k, c_m] = 0, \quad [c_k^\dagger, c_m^\dagger] = 0, \quad [c_k, c_m^\dagger] = \delta_{km}.$$

3) *Received Echo with Delay and Doppler Shift*: The received echo is:

$$G(t) = \alpha F\left(\kappa \left(t - \frac{2d}{c}\right)\right)$$

with Doppler factor $\kappa = 1 + v/c$.

The diagram illustrates the radar principle of delay and Doppler time scaling using a transmitted pulse and its received echo. The blue waveform on the left represents the transmitted signal pulse with duration (T_0). This pulse is emitted by the radar toward a target located at distance (d). After striking the target, the signal reflects back and returns to the radar after a round-trip travel time given by ($\tau = \frac{2d}{c}$), where (c) is the speed of light. This delay allows the radar to estimate the target range. The red waveform on the right is the received echo, which appears later in time because of this propagation delay. Its amplitude is reduced due to path loss and target reflection characteristics, hence it is labeled scaled. The echo duration becomes (κT_0), where (κ) is the Doppler scaling factor caused by target motion. If the target moves toward the radar, the pulse compresses; if it moves away, it stretches. Thus, delay gives range, while scaling reveals velocity.

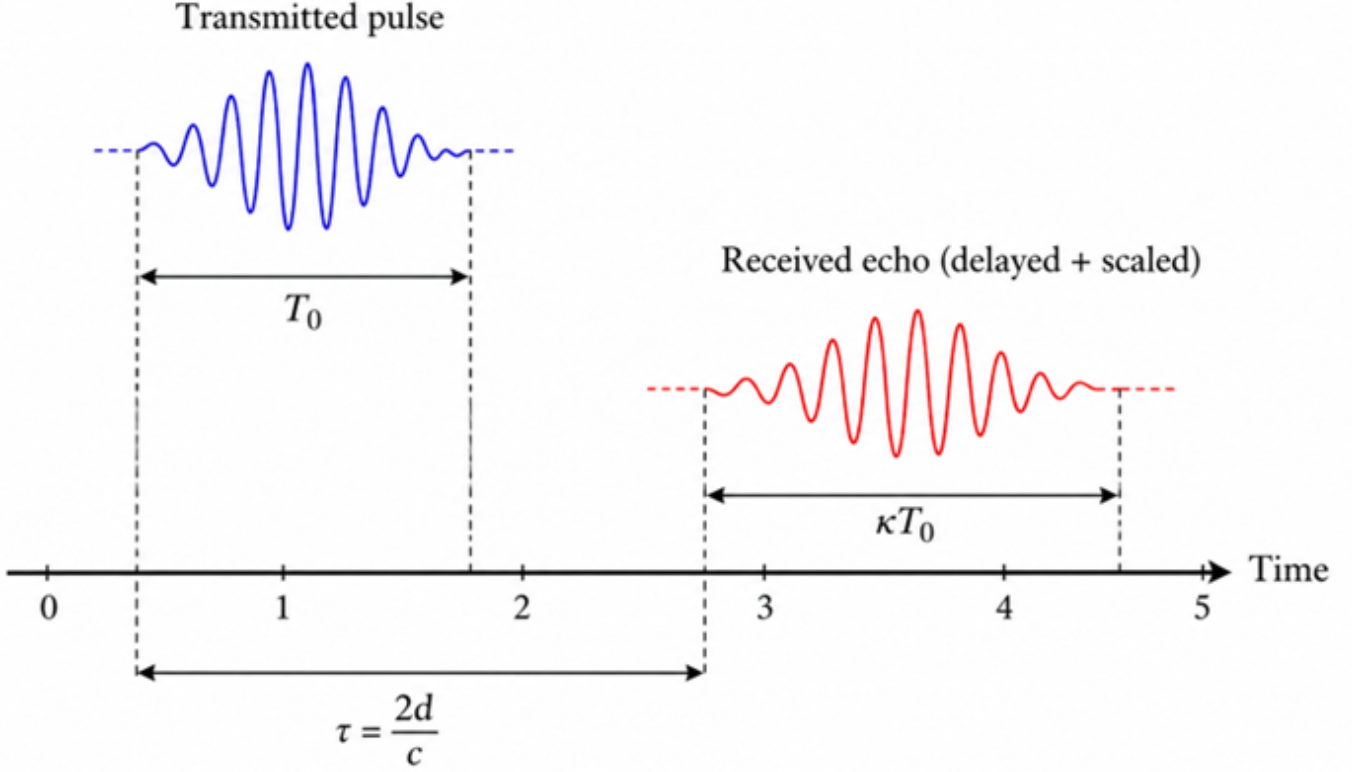


Fig. 2. Schematic of the radar principle of delay and Doppler time scaling using a transmitted pulse and its received echo.

4) Quantum Correlation Detector:

$$R_{GF}(t, \tau) = \text{Tr}(\rho G(t + \tau) F(t))$$

B. Multi-Target Quantum Antenna Model

The above diagram illustrates a multi-target sensing geometry for a classical radar, quantum radar, or advanced antenna system operating in a two-dimensional Cartesian coordinate plane. The horizontal axis represents the (x)-direction, while the vertical axis represents the (y)-direction, providing a spatial reference for locating the transmitter and multiple targets. The black point labeled "transmitter" at position ($\mathbf{r}_0$) denotes the radar or sensing node from which signals are emitted and echoes are received. The three red points correspond to separate targets located at distinct positions in space, such as ships, drones, vessels, aircraft, or moving objects. Dashed lines connect the transmitter to each target and represent the line-of-sight propagation paths. These paths are labeled (d_1), (d_2), and (d_3), which denote the distances or ranges between the transmitter and each respective target. Mathematically, each distance is expressed as the Euclidean norm between the target position vector and transmitter position, allowing determination of radar return delay through the relation ($\tau_k = 2d_k/c$), where (c) is the speed of light. Thus, farther targets generate longer return times. In addition to range, the geometry also enables estimation of target bearing angles based on the slope of each dashed path. Target 1 lies in the upper-right direction, Target 2 lies nearly horizontal to the transmitter, and Target 3 lies in the lower-right direction, showing how multiple

targets can be distinguished spatially. In a quantum radar framework, each target would additionally produce a unique reflected quantum signature characterized by range, angle, radial velocity, and scattering strength. The total received signal is a superposition of echoes from all targets plus environmental noise. This type of geometry forms the foundation of multi-target tracking, collision avoidance, coastal defense, maritime traffic monitoring, drone surveillance, and intelligent autonomous sensing systems. It is particularly valuable in future quantum-enhanced radar systems where simultaneous detection of weak or stealthy targets in cluttered environments is required.

For receiver position $\mathbf{r}_0$:

$$d_k(\mathbf{r}_0) = \|\mathbf{R}_k - \mathbf{r}_0\|, \quad \kappa_k(\mathbf{r}_0) = 1 + \frac{v_k(\mathbf{r}_0)}{c}$$

The multi-target received signal is:

$$G(t|\mathbf{r}_0) = \sum_{k=1}^p \frac{\alpha_k}{d_k} F\left(\kappa_k\left(t - \frac{2d_k}{c}\right)\right) + N(t|\mathbf{r}_0)$$

C. Entanglement-Assisted Quantum Radar

The diagram illustrates the operating principle of an entanglement-assisted quantum radar system, where quantum correlations between two generated beams are used to improve target detection in noisy environments. The left block labeled Entangled Source represents a quantum photon-pair generator that produces two strongly correlated signals: the signal beam (F_S) and the idler beam (F_I). These two beams are initially

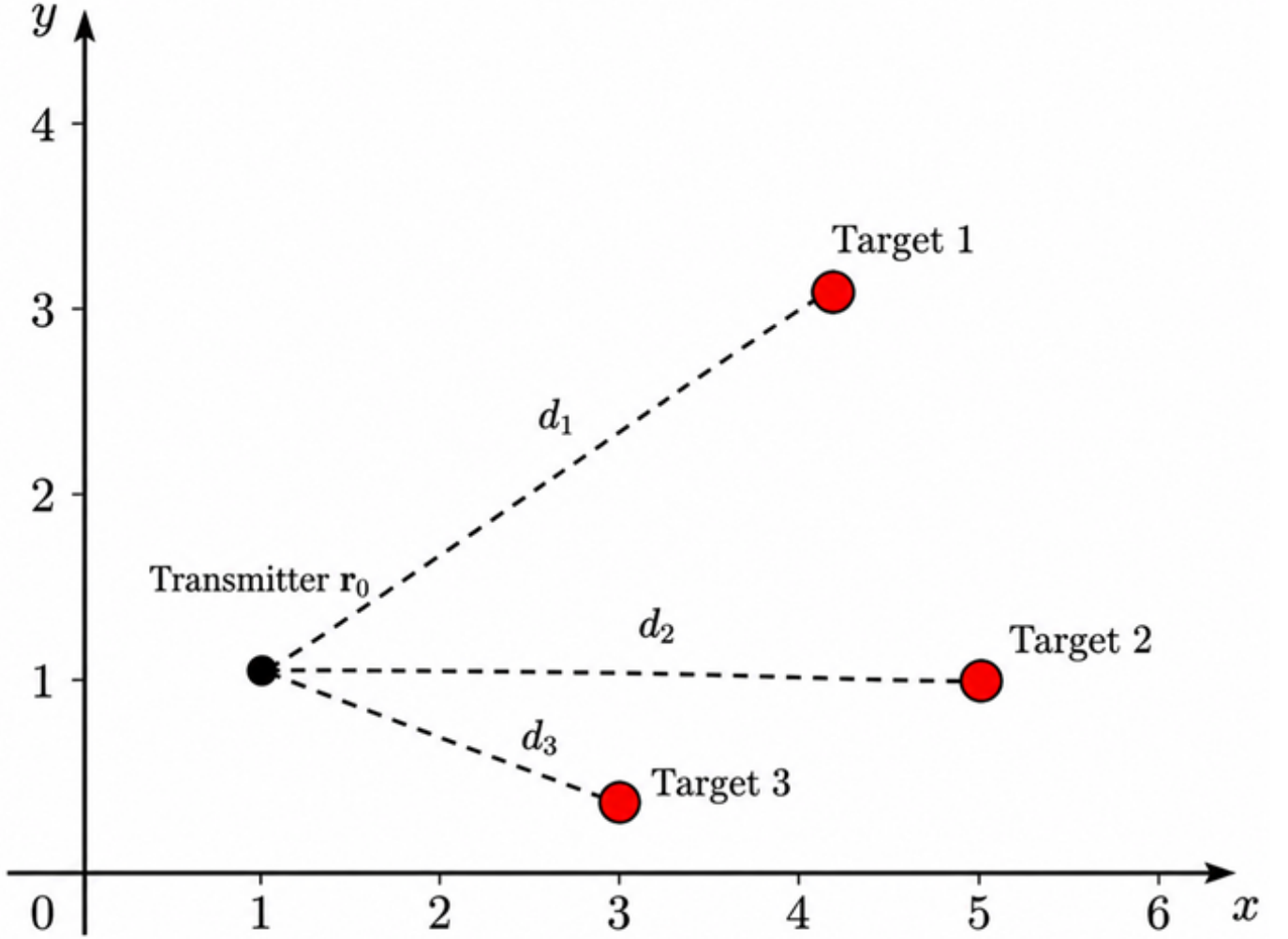


Fig. 3. Schematic of a multi-target sensing geometry quantum radar system

entangled, meaning their quantum properties remain linked even when separated. The signal beam (F_S) is transmitted toward the region containing the Target plus Channel, shown in the upper-right block. This block represents both the physical target and the surrounding propagation medium, including atmospheric loss, scattering, clutter, turbulence, and thermal noise. When the signal interacts with the target, a weak reflected echo is produced and travels back to the receiver. This returned signal is denoted by the dashed arrow labeled (G_S), indicating that the echo is attenuated, noisy, and may contain only a small amount of useful information. Meanwhile, the idler beam (F_I) is not transmitted. Instead, it is retained locally and routed directly to the Joint Detector. Because the idler has not passed through the noisy channel, it preserves the original quantum reference information created by the source. At the joint detector, the stored idler and the returned echo are measured together. Their residual correlations are processed to generate the correlation output, which reveals whether a target is present. Even if environmental noise destroys direct entanglement during propagation, the initial

quantum correlations can still provide a detection advantage over classical radar methods. This architecture is especially valuable for low-reflectivity targets, cluttered maritime environments, stealth object detection, and future high-sensitivity surveillance systems. The joint correlation is:

$$R_{SI}(t, \tau | \mathbf{r}_0) = \text{Tr} \left(\rho_{SI} G_S(t + \tau | \mathbf{r}_0) F_I(t) \right)$$

IV. QUANTUM ANTENNAS FROM THE SUPERCONDUCTIVITY CONTEXT

The antenna cavity consists of Fermions defined by the wave field $\psi_s(x)$, $s = \pm 1/2$ and $x = (t, r)$. Sometimes two Fermions can come together and form a Cooper pair defined by the wave field $\psi_{ss'}(x, x')$ as happens within a superconductor. Our aim is to calculate the Fermionic current density which drives the quantum mechanical radiation pattern. A quantum electromagnetic laser source is inserted into the cavity at a certain point thereby causing the Fermions to interact with the photons thereby altering the Fermionic current that radiates out into space. The second quantized action of the Fermions

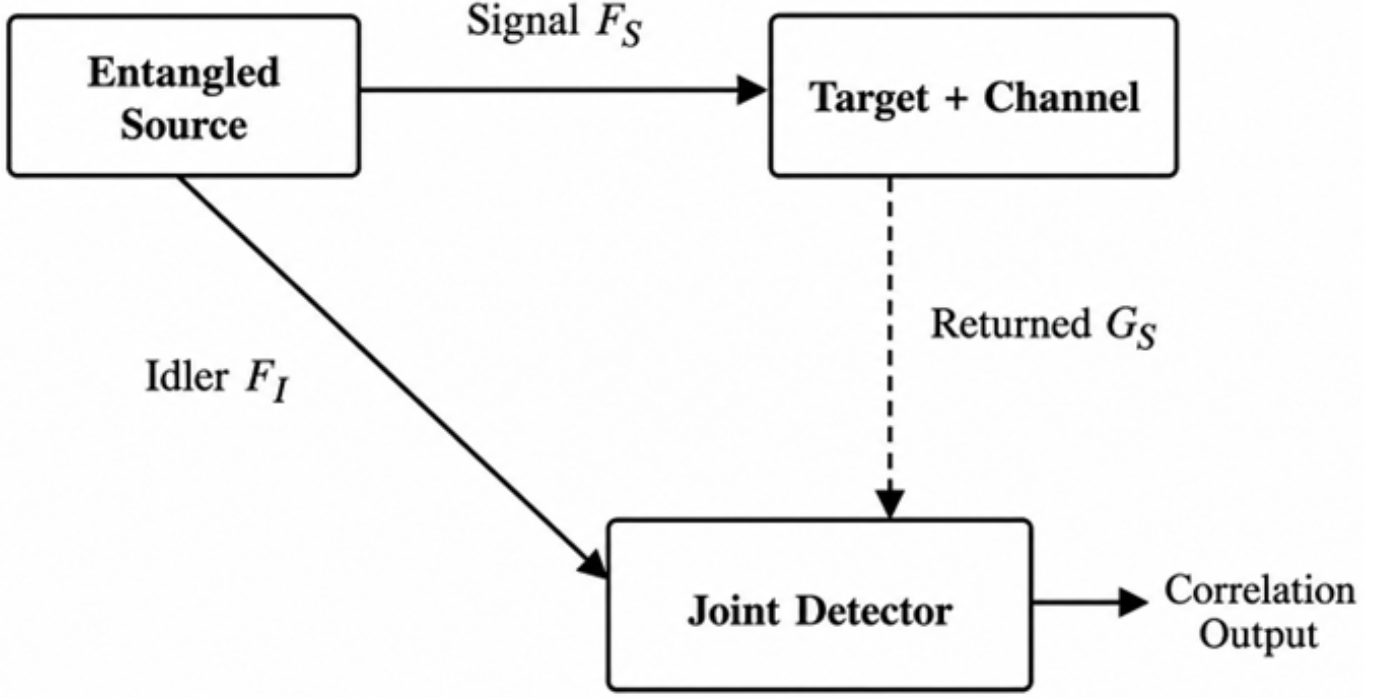


Fig. 4. Schematic of an entanglement-assisted quantum radar system

taking into account its interactions with the photons is given by

$$S[\psi, \psi^*] = \int \psi_s(x)^* (i\hbar \partial_t - E(-i\hbar \nabla + eA(x)) + eV(x)) \psi_s(x) d^4x + \int V_{ss'}(x, y) \psi_s(x)^* \psi_s(x) \psi_{s'}(y)^* \psi_{s'}(y) d^4x d^4y$$

where $V(x)$ is the scalar potential and $A(x)$ the magnetic vector potential associate with the electromagnetic field. For the present purposes we assume the electromagnetic field to be a classical field because we shall be using it to control the dynamics of the electron field and hence the electronic current. The second term that is quartic in the Fermionic wave field represents the interaction energy between two Cooper pairs. More precisely, $\psi_s(x)^* \psi_s(x)$ equals the number density operator of Fermions at space-time location x having z-component of spin quantum number s and likewise for $\psi_{s'}(y)^* \psi_{s'}(y)$. Then, $V_{ss'}(x, y)$ represents the interaction potential energy between one Fermion of spin polarization s at x and another Fermion of spin polarization s' at y , and consequently, the quartic integral represents the interaction term between Fermions of same or different polarizations located at two different space time points wherein the interaction is assumed to be proportional to the number of pairs that we can form using Fermions at two different locations, namely, the product of the number of Fermions at the two different space-time points. It should be noted that E stands for the free Dirac Hamiltonian:

$$E(P) = (\alpha, P) + \beta m$$

so that

$$E(-i\hbar \nabla + eA) - eV = (\alpha, -i\hbar \nabla + eA) + \beta m - eV$$

is the Dirac Hamiltonian of a Fermion taking into account its interaction with the electromagnetic field. Using this Fermionic action, we can introduce a Cooper pair field $\psi_{ss'}(x, y)$ by adding to the action S , the term

$$\Delta S = - \int V_{ss'}(x, y) (\psi_{ss'}(x, y) - \psi_s(x) \psi_{s'}(y)) (\psi_{ss'}(x, y)^* - \psi_s(x)^* \psi_{s'}(y)^*) d^4x d^4y$$

and considering the path integral

$$\int \exp(i(S + \Delta S)/\hbar) D\psi_s D\psi_{s'} D\psi_{ss'} D\psi_{ss'}^*$$

The addition of ΔS will not change the value of the path integral because the integral over $D\psi_{ss'}, D\psi_{ss'}^*$ removes its effect by replacing the path integral for S by a multiplicative normalization constant only. The quantum effective action for the pair field is then given by

$$\exp(i\Gamma_{pair}[\psi_{ss'}, \psi_{ss'}^* | A, V]) = \int \exp(i(S + \Delta S)/\hbar) D\psi_s D\psi_s^*$$

and this quantum effective action can be used to calculate the Josephson current to a very high degree of accuracy. This Josephson current is a purely quantum mechanical effect and is of a very high frequency, because it is proportional to $1/\hbar$ where $\hbar$ is Planck's constant. By controlling the electromagnetic field A, V , we can also control the amplitude

of the Josephson current and this current can be used as a very high frequency range transmitter antenna.

V. QUANTUM RADAR USING ENTANGLED STATES

The initial state of the Tx and Rx is $\rho_T(0) \otimes \rho_R(0)$. Let H_I denote the interaction Hamiltonian between the Tx and Rx fields. Then, after evolving for time T , the joint state of Tx and Rx becomes entangled:

$$\rho(\text{ent}) = U(T)(\rho_T(0) \otimes \rho_R(0))U(T)^*$$

We can use this shared entangled state to teleport a state $\rho(1)$ from Tx to Rx by adjoining this state to $\rho(\text{ent})$ on the Tx side, making measurements and conveying the result of these measurements through a classical channel between Tx and Rx. Of course, the interaction Hamiltonian H_I between the fields of Tx and Rx depend upon the relative location of Rx w.r.t Tx and also on the velocity of Rx relative to Tx. Hence, the entangled state $\rho(\text{ent})$ is also a function of these parameters and the teleported state will therefore have some distortion. After Rx receives this distorted state, he again uses his shared entanglement with Tx to reverse teleport this distorted state to Tx. Tx will therefore finally receive a distorted version of the state that he had originally intended to transmit. In fact this distortion would be a double distortion, one during teleportation from Tx to Rx and two, during teleportation from Rx to Tx. By measuring this distorted state, Tx can infer the parameters of Rx relative to Tx, namely, relative range, bearing and velocity. This is our proposed version of quantum radar based on entangled states. Here, the reflection by the target back to the transmitter actually corresponds to reverse teleportation. Once Tx gets to know of these parameters of Rx using quantum entanglement, he can reorient his quantum antenna to transmit and receive messages from Rx.

VI. CONCLUSION

In this paper, we have introduced a novel framework for quantum radar based on the dynamic generation and utilization of entangled quantum states between the transmitter (Tx) and the target/receiver (Rx). Starting from an initial separable state $\rho_T(0) \otimes \rho_R(0)$, the bilinear interaction Hamiltonian H_I governing the electromagnetic field coupling between Tx and Rx naturally evolves the joint system into an entangled state after time T :

$$\rho_{\text{ent}} = U(T) [\rho_T(0) \otimes \rho_R(0)] U^\dagger(T),$$

where

$$U(T) = \mathcal{T} \exp \left(-i \int_0^T H_I(t) dt \right)$$

is the time-evolution operator. This entangled state is parametrically dependent on the relative range vector, radial velocity, angular bearing, and rotational dynamics of the target, as these parameters directly influence the interaction strength, phase accumulation, and Doppler shifts encoded in H_I . The proposed quantum radar operates through a bidirectional time delay involved in the exchange of signals between the

the transmitter teleports a chosen quantum state $\rho(1)$ to the target. Upon interaction with the target, the received state undergoes a dependent unitary evolution. This distorted state is then reverse-teleported back to the transmitter via the same entangled resource. The f

By performing sequential measurements on the returned distorted state and applying appropriate quantum estimation techniques (such as quantum hypothesis testing, maximum likelihood estimation, or Bayesian inference on the entangled resource), the transmitter can infer the target's kinematic parameters — range, velocity, and orientation — with precision potentially surpassing the standard quantum limit. The reflection process from the target is effectively modeled as a reverse teleportation channel, thereby unifying the radar echo mechanism with quantum information protocols.

This approach distinguishes itself from conventional quantum illumination and microwave quantum radar schemes by explicitly exploiting bidirectional entanglement-mediated teleportation rather than relying solely on signal-idler correlations or photon counting. It naturally integrates with advanced entangled photon sources such as Graphene Quantum Dots (GQDs) operating via biexciton cascade emission, ion-trap assisted coherence preservation, and rubidium-enhanced microwave sensing, as outlined in the hybrid architecture.

The framework also incorporates rigorous theoretical foundations, including the Born scattering approximation and Feynman path integral formalism for modeling photon-target interactions, enabling physically consistent simulation of quantum interference effects across all possible propagation paths.

Future Directions include the following:

1. Derivation of the Quantum Fisher Information (QFI) and Cramér-Rao bound for multiparameter estimation of range, velocity, and angle.
2. Optimization of the interaction Hamiltonian H_I for maximal parameter encoding.
3. Experimental realization using GQD-based entangled photon pairs in the microwave regime.
4. Integration with adaptive quantum antenna arrays for real-time beam steering based on extracted parameters.
5. Analysis of robustness against decoherence, loss, and jamming using quantum error correction and dynamical decoupling techniques.

In summary, the proposed quantum radar paradigm based on entangled-state teleportation and bidirectional distortion analysis offers a powerful new pathway for achieving high-precision, stealth-resistant target detection and tracking. By bridging quantum information science with radar engineering, this work contributes to the advancement of next-generation quantum sensing technologies with significant strategic applications in defense, surveillance, and autonomous systems.

APPENDIX A

Here we introduce some practically useful generalizations of the quantum radar problem discussed in the main body of the paper.

It involves considering the target as a 3-D rigid body moving in space with a range-bearing vector at time t being $d(t) \in \mathbb{R}^3$, a velocity vector at time t being $v(t) \in \mathbb{R}^3$, a rotation $R(t) \in SO(3)$ of the target within its own frame of reference and a time delay involved in the exchange of signals between the

Tx and the target. In this, the velocity target parameters to be estimated on a real time basis is given by

$$\Theta(t) = (\tau(t), d(t), v(t), R(t)) \in \mathbb{R} \times \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3)$$

which comprises the Poincaré group as the semi-direct product of space-time translations with the Lorentz group of boosts and rotations.

If $E_T(x)$, $x = (t, r)$, denote the Tx EM field

$$\begin{bmatrix} E_T(x) \\ B_T(x) \end{bmatrix}$$

and $F'_R(x')$ denotes the EM field of both the targets in their own rest frames, in K , K' respectively.

Transforming the target field into the Tx frame using the Poincaré transformations gives

$$F_R(x | \theta(t)) = U \left(T_{\theta(t)}, F'_R(T_{\theta(t)}x) \right) \quad (1)$$

where U denotes the 4×4 representation of the Poincaré transformation group.

The total Hamiltonian in Tx and target is

$$H(t) = Q(F_T + F_R) \quad (2)$$

$$= Q(F_T) + Q(F_R) + B(F_T, F_R) \quad (3)$$

where Q is the quadratic energy form and $B(\cdot, \cdot)$ the induced symmetric bilinear form.

The interaction Hamiltonian is

$$\begin{aligned} H_I(t) &= B(F_T, F_R) \\ &= \int (F_T(x), Q_0 F_R(x | \Theta(t))) d^3x \\ &= H_I(\Theta(t)) \end{aligned}$$

where Q_0 is the medium matrix determined by the permittivity and permeability tensor of the medium.

We can write

$$\begin{aligned} F_T(x) &= \sum_k c_k^T \chi_{kT}(x) + c_k^{T\dagger} \bar{\chi}_{kT}(x) \\ F_R(x') &= \sum_k c_k^R \chi_{kR}(x') + c_k^{R\dagger} \bar{\chi}_{kR}(x') \end{aligned}$$

where $c_k^T, c_k^{T\dagger}$, $k = 1, 2, \dots$

are photon annihilation-creation operators in the Tx field and

$$(c_k^R, c_k^{R\dagger}), \quad k = 1, 2, \dots$$

are photon annihilation-creation operators of the targets.

The entangled states between the Tx and target evolves as

$$U(t)(|\psi_T(0)\rangle \otimes |\psi_R(0)\rangle)$$

where

$$U(t) = \mathcal{T} \left\{ \exp \left(-i \int_0^t H_B(\Theta(u)) du \right) \right\}$$

These entangled states can be estimated by measurements as the Tx and all the targets at time $\Theta(t)$ can be estimated on a real time basis.

REFERENCES

- [1] Mario Agio and Andrea Alù. Optical antennas. Cambridge University Press, 2013. VOLUME, This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see <https://creativecommons.org/licenses/by/4.0/>This article has been accepted for publication in IEEE Open Journal of Antennas and Propagation. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/OJAP.2025.3545596
- [2] Yoshihisa Yamamoto, Francesco Tassone, and Hui Cao. Semiconductor cavity quantum electrodynamics, volume 169. Springer, 2003.
- [3] A Blais, RS Huang, A Wallraff, SM Girvin, and RJ Schoelkopf. Cavity quantum electrodynamics for superconducting electrical circuits: An architecture for quantum computation. *Physical Review A*, 69(6):062320, 2004.
- [4] Wilfred G Van der Wiel, Silvano De Franceschi, Jeroen M Elzerman, Toshimasa Fujisawa, Seigo Tarucha, and Leo P Kouwenhoven. Electron transport through double quantum dots. *Reviews of modern physics*, 75(1):1, 2002.
- [5] Robert L Olmon and Markus B Raschke. Antenna-load interactions at optical frequencies: impedance matching to quantum systems. *Nanotechnology*, 23(44):444001, 2012.
- [6] Alexandra S Sheremet, Mihail I Petrov, Ivan V Iorsh, Alexander V Poshakinskiy, and Alexander N Poddubny. Waveguide quantum electrodynamics: collective radiance and photon-photon correlations. *Reviews of Modern Physics*, 95(1):015002, 2023.
- [7] Lorenzo Leandro, Jacob Hastrup, Rodion Reznik, George Cirlin, and Nika Akopian. Resonant excitation of nanowire quantum dots. *npj Quantum Information*, 6(1):93, 2020.
- [8] Toichiro Kinoshita. Quantum electrodynamics, volume 7. World Scientific, 1990.
- [9] Ryota Katsumi, Yasutomo Ota, and Mohamed Benyoucef. Telecomband quantum dots compatible with silicon photonics for photonic quantum applications. *Advanced Quantum Technologies*, page 2300423, 2024.
- [10] P. Lodahl, S. Mahmoodian, and S. Stobbe, "Interfacing single photons and single quantum dots with photonic nanostructures," *Rev. Mod. Phys.*, vol. 87, no. 2, pp. 347–400, 2015.
- [11] S. Buckley, K. Rivoire, and J. Vuckovic, "Engineered quantum dot single-photon sources," *Rep. Prog. Phys.*, vol. 75, no. 12, 126503, 2012.